AWS IAM Users, Permissions and Policies

Practical Documentation

1. Introduction

AWS Identity and Access Management (IAM) is used to control access to AWS resources by managing users, permissions, and policies.

This practical covers creating an IAM user, testing access to an EC2 instance, assigning permissions, exploring IAM policies, creating a custom EC2 policy, and using the AWS Policy Generator to understand IAM policy structure.

2. Objective

To understand how AWS IAM manages access by creating a user, testing access to an EC2 instance, assigning permissions, and creating and examining IAM policies.

3. Topology

The topology shows the complete flow followed in this practical, starting with IAM user creation and ending with IAM policy generation.

The Root/Admin user first creates an IAM user and provides the required console sign-in details. At this stage, the IAM user does not have the permissions required to access the EC2 resources.

The Root/Admin then launches an EC2 instance that is used for testing. The IAM user signs in to the AWS Management Console and tries to access EC2. Since the required permissions are not yet attached, AWS denies access to several EC2 resources.

The Root/Admin then returns to IAM and adds a policy to the IAM user. After the permission is attached, the IAM user accesses EC2 again and the instance becomes available.

The practical then moves to IAM policy management. A custom EC2 policy is created by selecting actions such as `StartInstances`, `StopInstances`, and `TerminateInstances`. The EC2 resource is specified using its Amazon Resource Name (ARN).

Finally, the AWS Policy Generator is used to create an IAM policy statement by selecting the effect, EC2 service, required actions, and resource. The generated policy is then displayed in JSON format.

Overall, the topology connects the Root/Admin, IAM user, IAM service, policies, EC2 instance, and Policy Generator to show how IAM permissions control access to AWS resources.

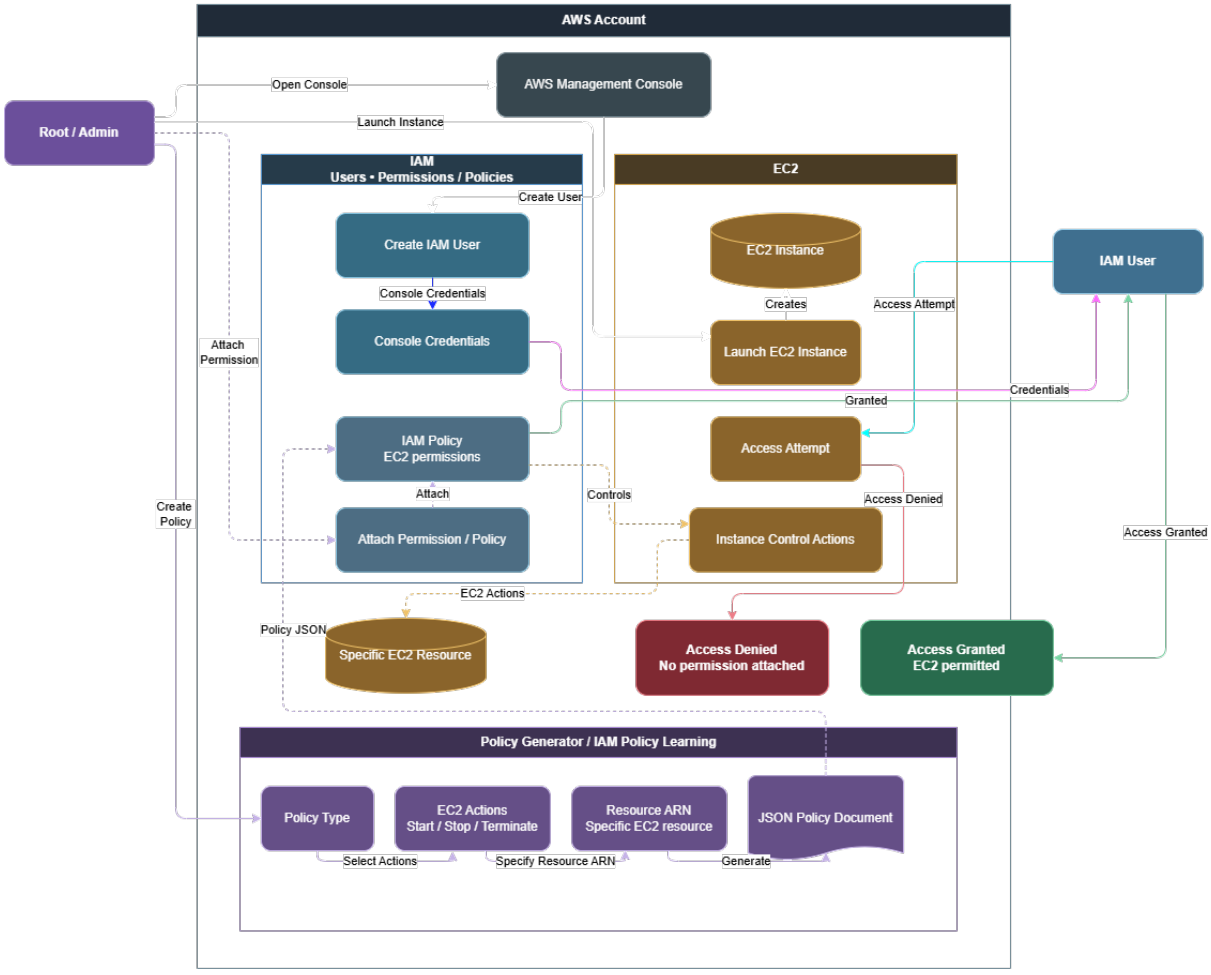


Figure 1: AWS IAM and EC2 Access Topology

4. Steps

4.1. Step 1: Open IAM Users

First, log in to the AWS Management Console and open the IAM service.

Go to the **Users** section. The IAM Users page appears with the option to create a new user.

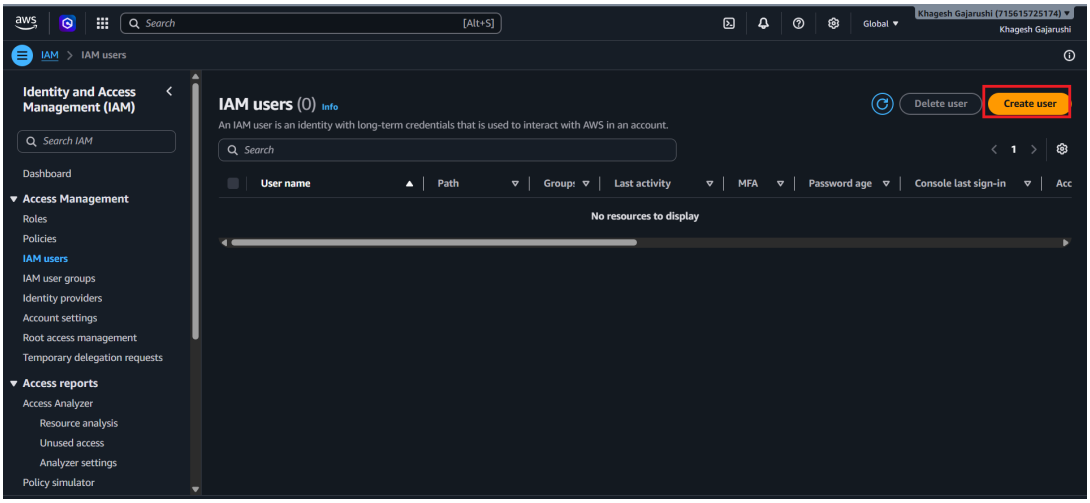


Figure 2: IAM Users page

4.2. Step 2: Create the IAM User

Select **Create user** and enter the required username.

Enable AWS Management Console access and configure a custom password for the new IAM user.

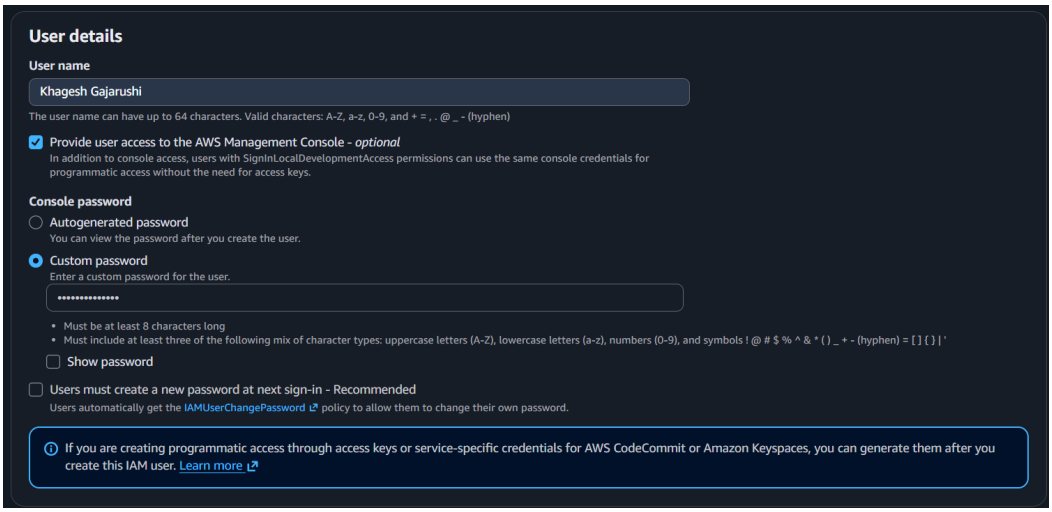


Figure 3: IAM user details and console access configuration

4.3. Step 3: Configure Permissions

Move to the permissions section while creating the user.

The option to attach policies directly to the user is selected.

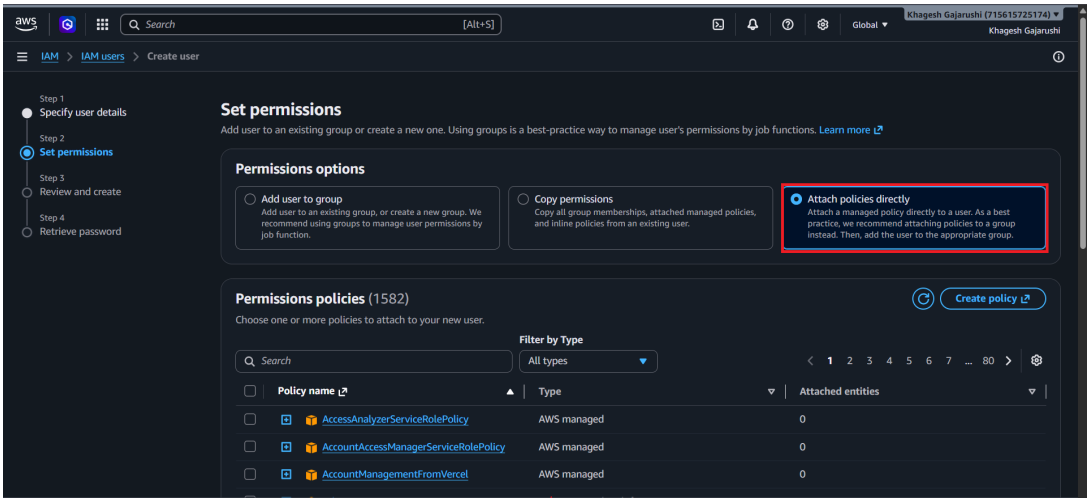


Figure 4: IAM user permission configuration

4.4. Step 4: Review the User

Review the IAM user details and permission settings before creating the user.

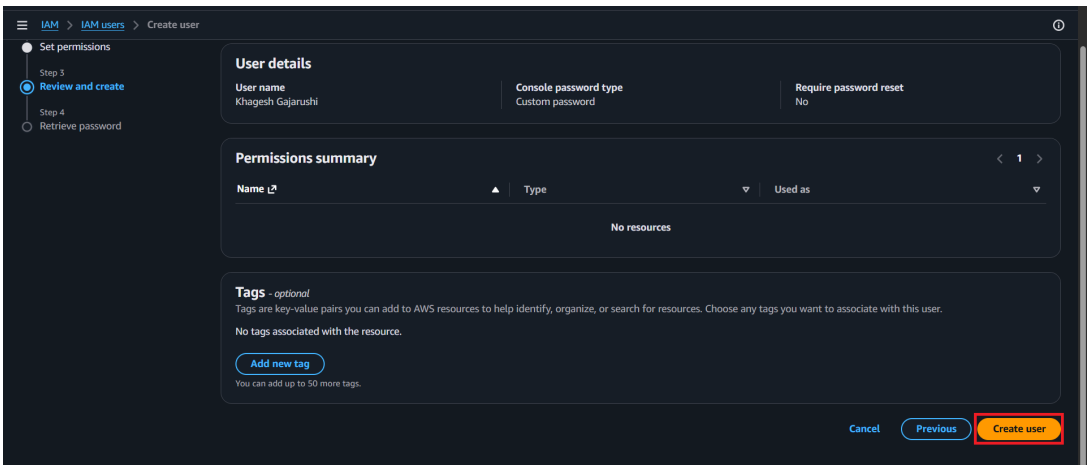


Figure 5: Reviewing IAM user configuration

4.5. Step 5: Create the User

Create the IAM user.

AWS displays a successful creation message and provides the option to retrieve the user's sign-in information.

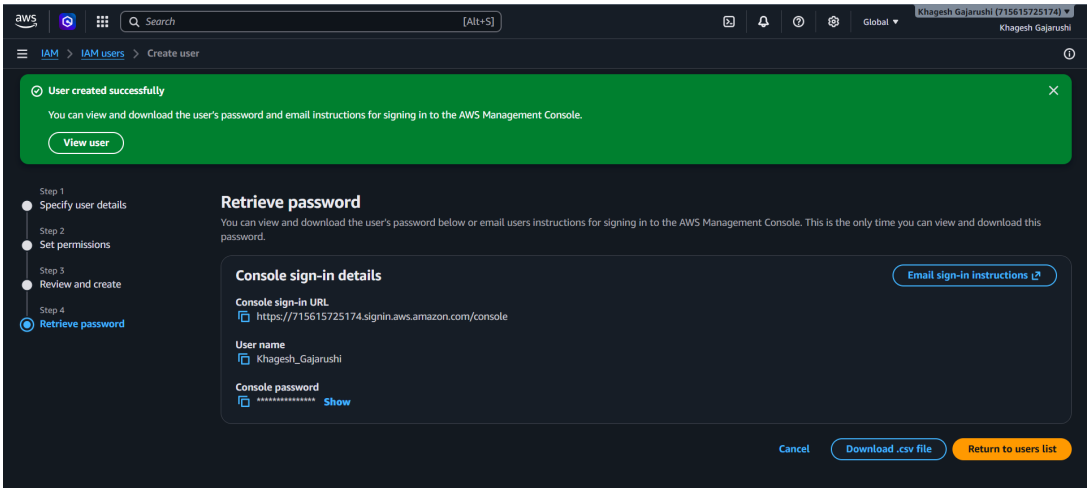


Figure 6: Successful IAM user creation

4.6. Step 6: Obtain Account Information

The AWS account ID is needed when signing in as the IAM user.

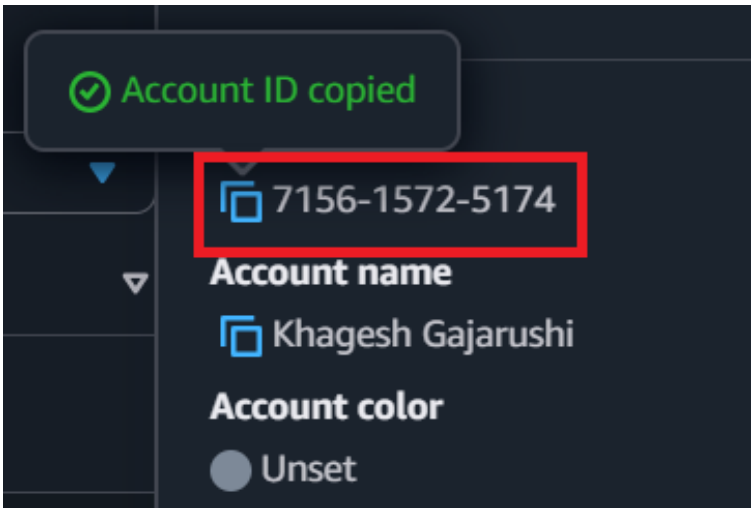


Figure 7: AWS account information

4.7. Step 7: Sign In as the IAM User

Use the AWS account ID, IAM username, and password to sign in to the AWS Management Console as the newly created IAM user.

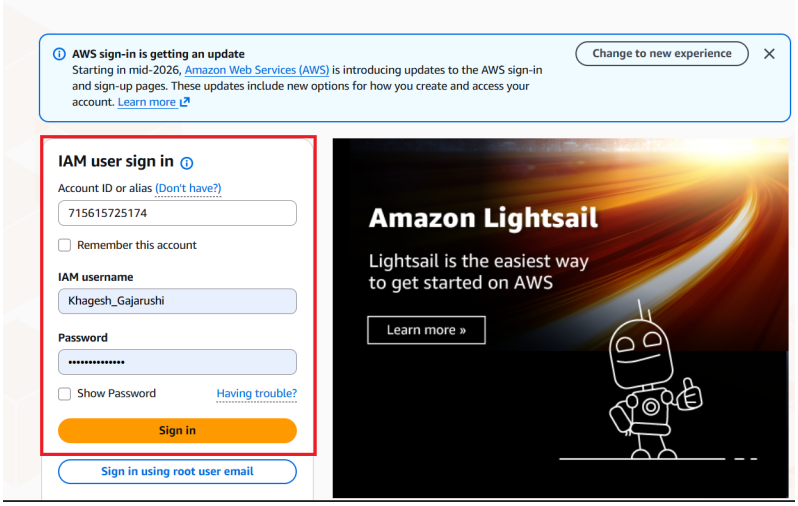


Figure 8: IAM user sign-in

4.8. Step 8: Launch an EC2 Instance

Return to the Root/Admin account and open the EC2 service.

Launch an EC2 instance using the required configuration. The instance used in this practical is the `IAM demo` instance.

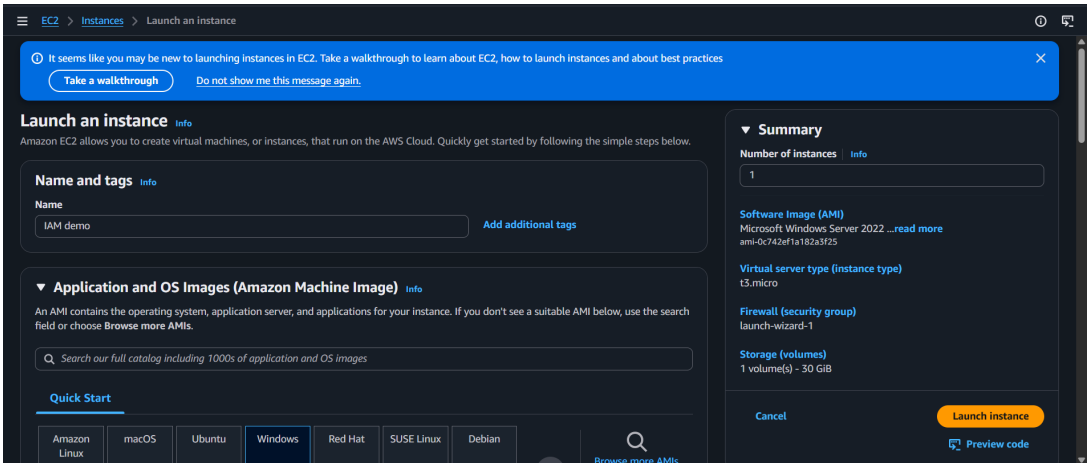


Figure 9: EC2 instance launch configuration

4.9. Step 9: Verify the EC2 Launch

After launching the instance, AWS displays a successful launch confirmation.

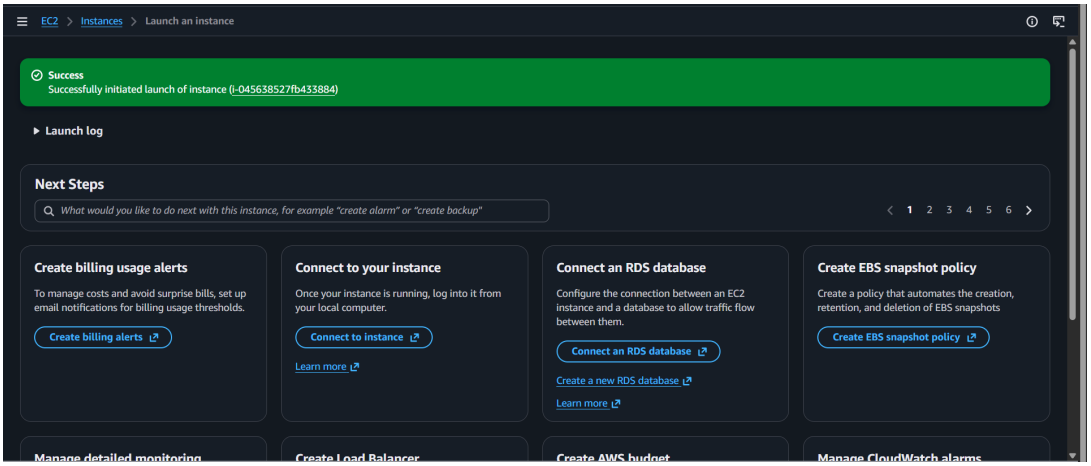


Figure 10: Successful EC2 instance launch

4.10. Step 10: Test EC2 Access as IAM User

Switch back to the IAM user and open the EC2 service.

The user is unable to access several EC2 resources, and AWS displays **Access denied** messages because the required permissions have not yet been provided.

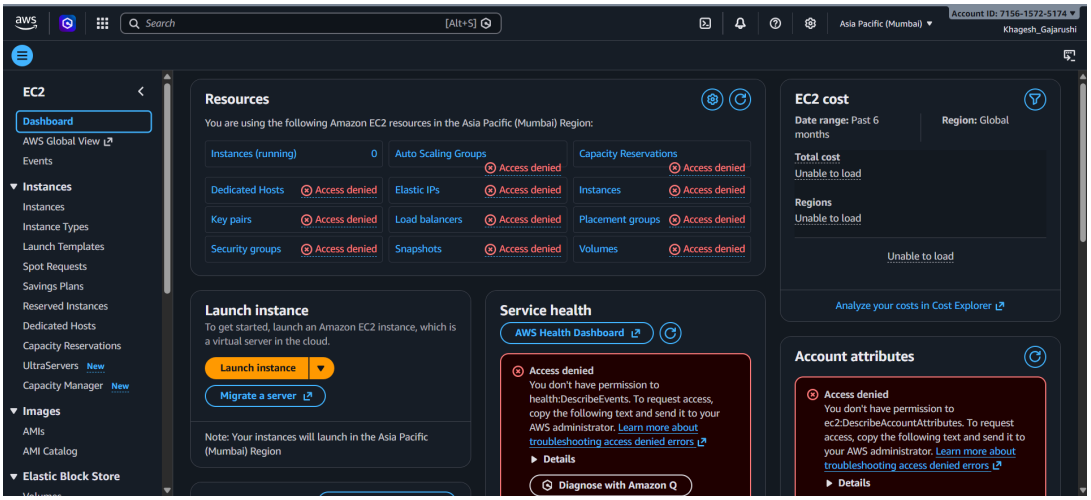


Figure 11: EC2 access denied for the IAM user

4.11. Step 11: Check IAM User Permissions

Return to the Root/Admin account and check the IAM user's permissions.

At this point, the user has no permission policies attached.

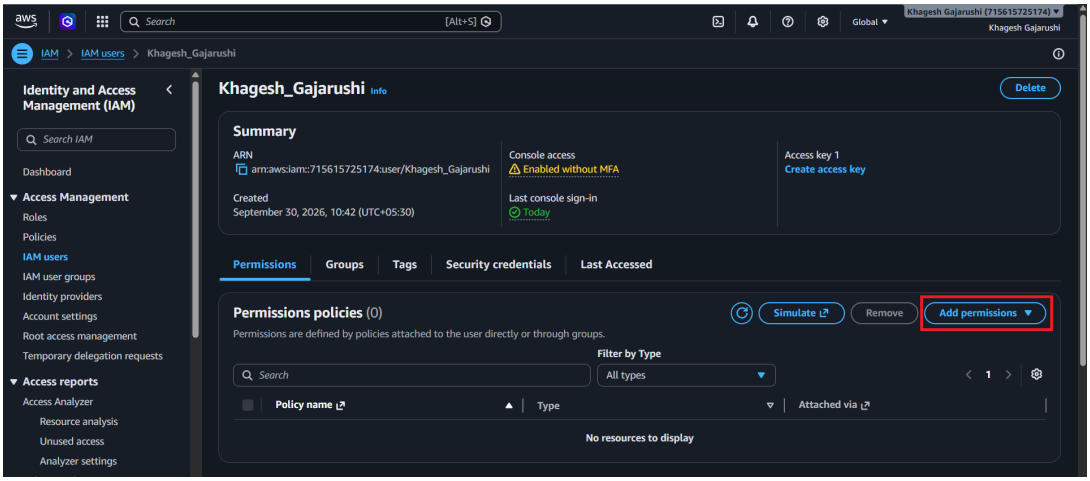


Figure 12: IAM user with no permission policies

4.12. Step 12: Add Permissions

Select **Add permissions** for the IAM user.

Choose the option to attach policies directly to the user and select the required policy.

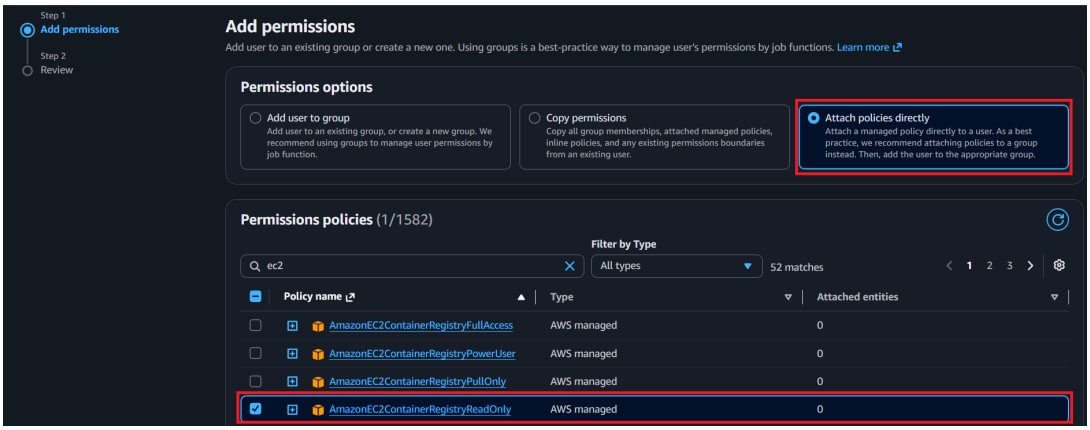


Figure 13: Adding permissions to the IAM user

4.13. Step 13: Review and Attach the Policy

After selecting the policy, AWS displays a review page showing the policy that will be attached to the IAM user.

Review the permission details and confirm the configuration.

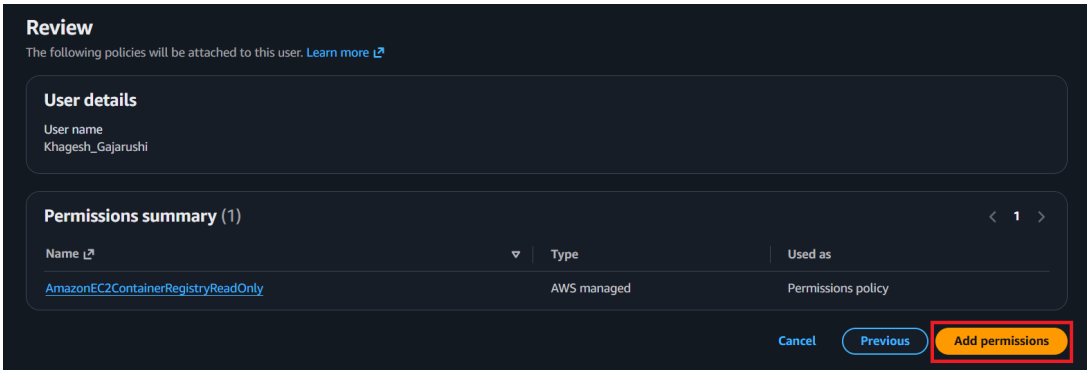


Figure 14: Reviewing the selected IAM policy

4.14. Step 14: Verify the Permission

Return to the IAM user and open EC2 again.
The EC2 instance is now visible to the IAM user after the permission change.

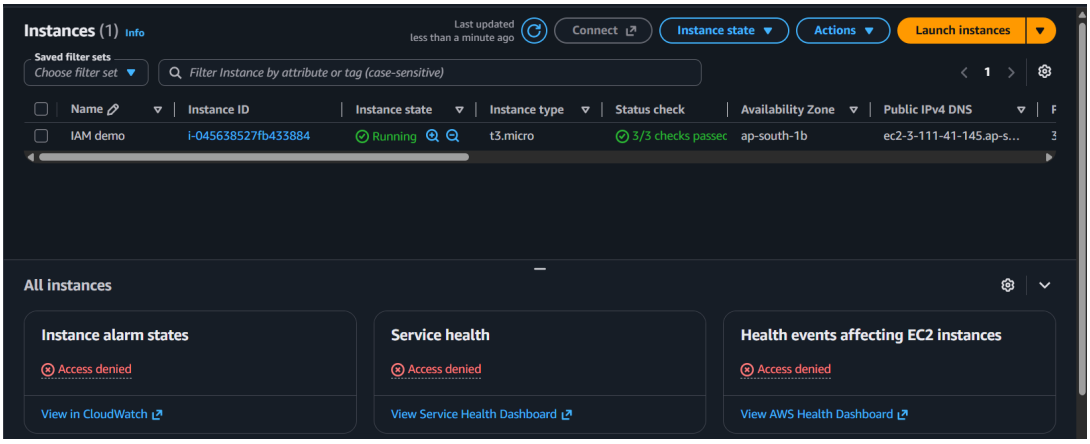


Figure 15: EC2 instance accessible after permission change

4.15. Step 15: Explore IAM Policies

Return to the Root/Admin account and open the IAM **Policies** section.
The available policies can be viewed and explored from this page.

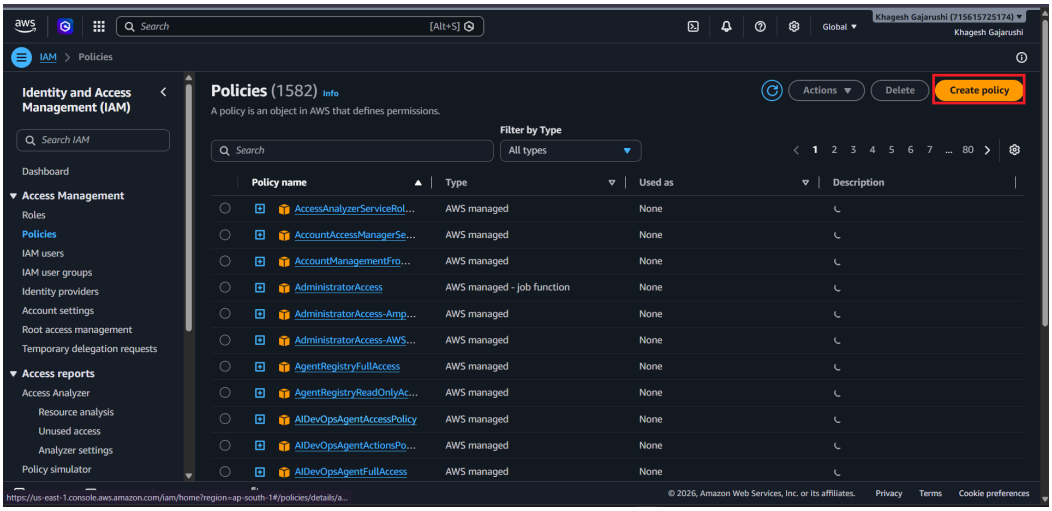


Figure 16: IAM Policies section

4.16. Step 16: Select EC2 Actions

Create a custom policy and select Amazon EC2 as the service.
Choose the EC2 actions required for the policy, including:

- `StartInstances`
- `StopInstances`
- `TerminateInstances`

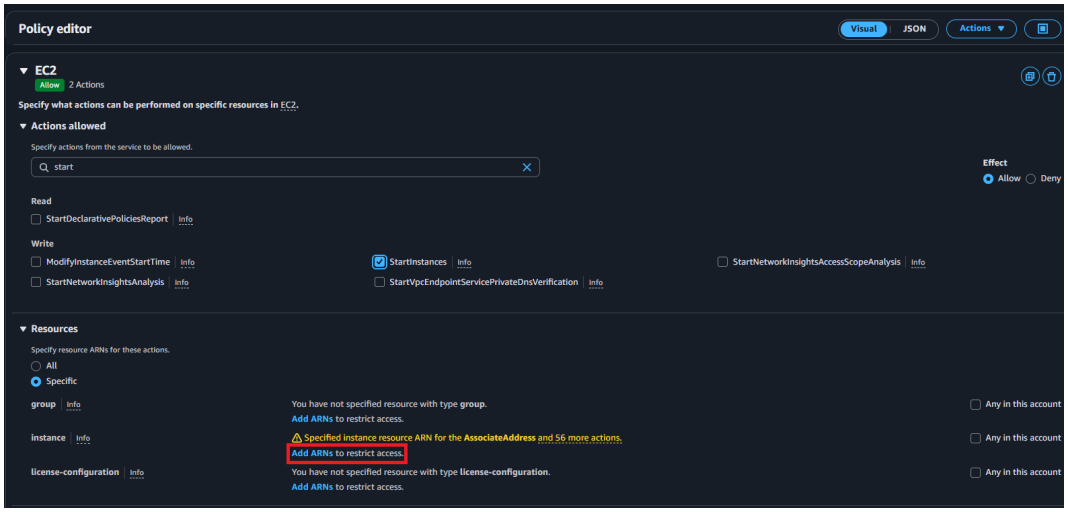


Figure 17: Selecting EC2 actions for the policy

4.17. Step 17: Specify the EC2 Resource

Specify the EC2 resource using its Amazon Resource Name (ARN).
The resource is associated with the EC2 instance in the `ap-south-1` region.

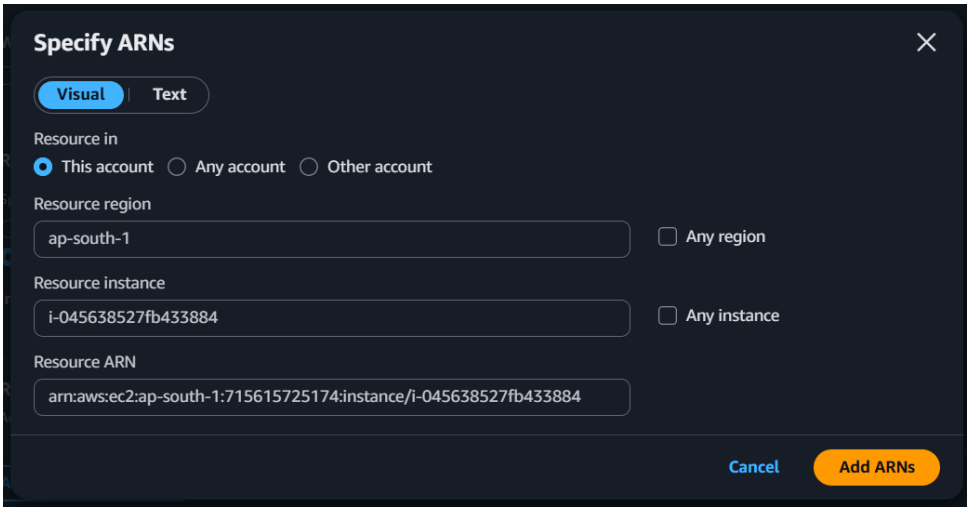


Figure 18: Specifying the EC2 resource ARN

4.18. Step 18: Create the Custom Policy

Complete the policy configuration and create the custom EC2 policy.
The newly created policy appears in the IAM Policies section.

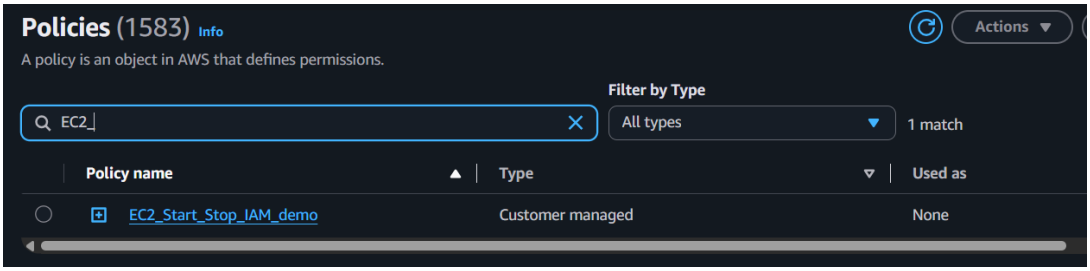


Figure 19: Custom EC2 policy

4.19. Step 19: Test EC2 Instance Control

Use the configured permissions to perform an EC2 instance operation.
The stop operation is initiated successfully for the EC2 instance.

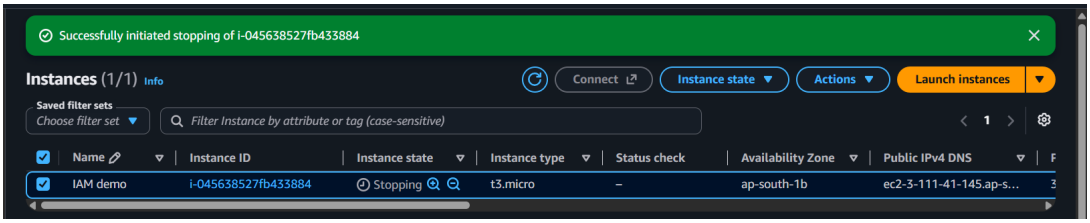


Figure 20: EC2 instance stop operation initiated

4.20. Step 20: Verify the EC2 Instance

Check the EC2 instance again after performing the stop operation.
The instance status is displayed in the EC2 console.

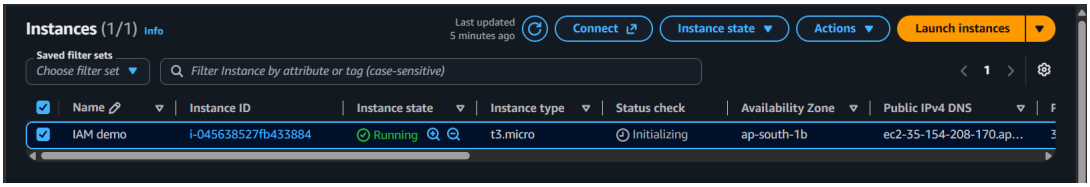


Figure 21: EC2 instance after the control operation

4.21. Step 21: Open AWS Policy Generator

Open the AWS Policy Generator and select **IAM Policy** as the policy type.

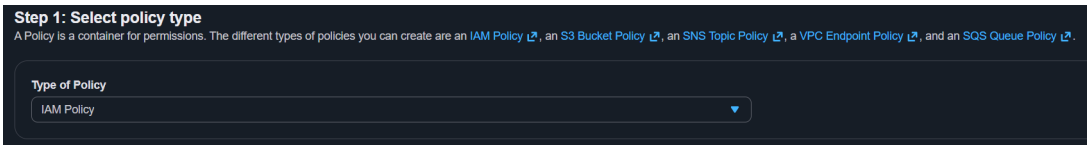


Figure 22: Selecting IAM Policy in AWS Policy Generator

4.22. Step 22: Configure the Policy Statement

Add the policy statement by selecting the required effect, Amazon EC2 service, actions, and resource.
The configured statement is added to the policy.

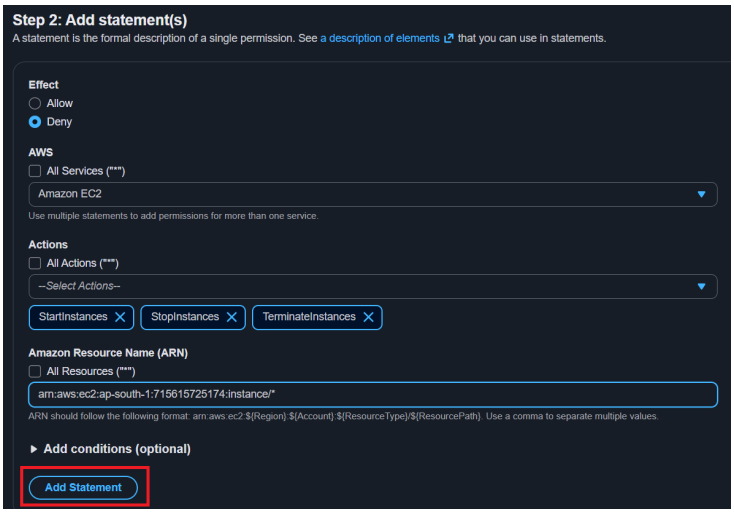


Figure 23: Configuring the IAM policy statement

4.23. Step 23: Generate the Policy

Review the configured policy statement and select **Generate Policy**.

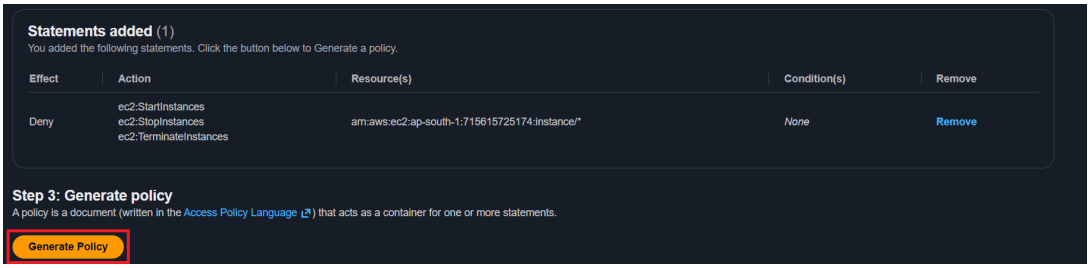


Figure 24: Generating the IAM policy

4.24. Step 24: View the Policy JSON

The generated policy is displayed as a JSON document.

The JSON contains the main IAM policy elements, including the effect, actions, and resource.

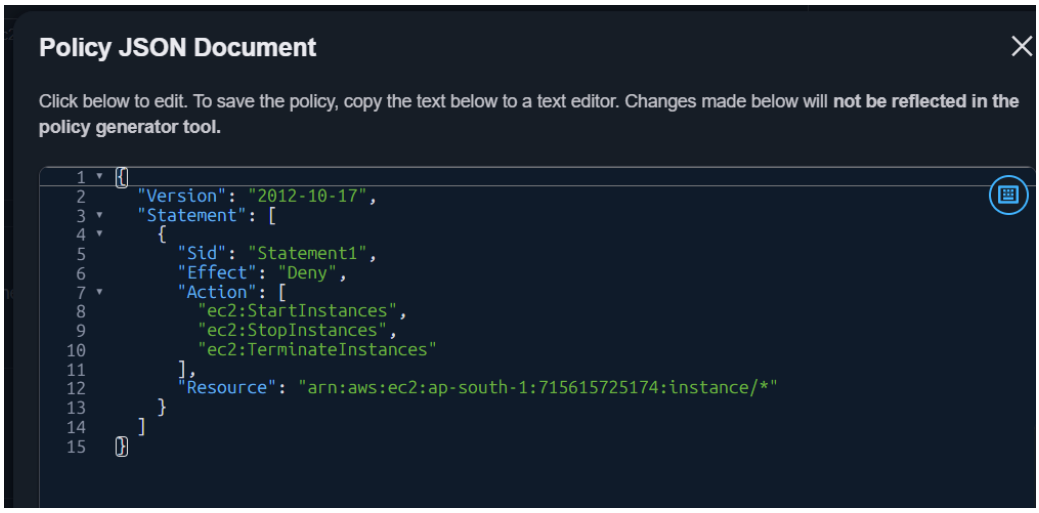


Figure 25: Generated IAM policy JSON

5. Conclusion

The practical demonstrated how AWS IAM controls access to AWS resources through users, permissions, and policies. An IAM user was created and tested against an EC2 instance, initially without the required permissions. Permissions were then assigned, and the change in access was observed.

The practical was further extended by exploring IAM policies, creating a custom EC2 policy, testing EC2 operations, and using the AWS Policy Generator to understand IAM policy JSON.